Name: _____

WORKSHEET

The inclusion-exclusion principle

1 For each of these sets, find:

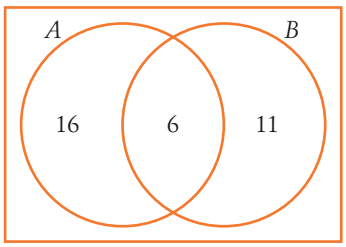
i $|A|$

ii $|B|$

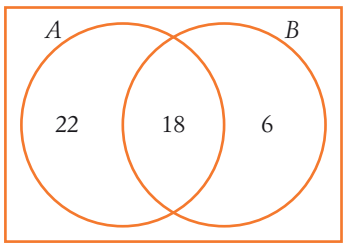
iii $|A \cap B|$

iv $|A \cup B|$

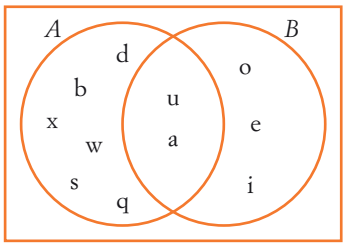
a



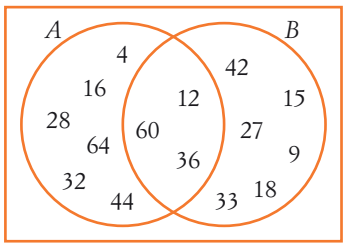
b



c

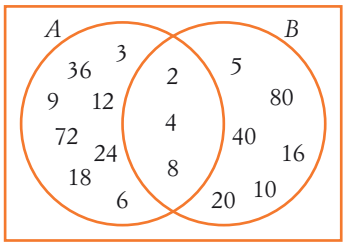


d

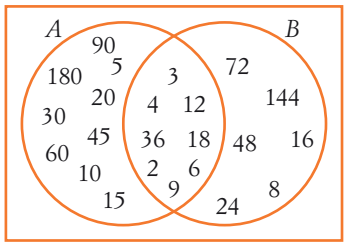


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e



f



2 For each of these sets, find:

i $|A|$

ii $|B|$

iii $|A \cap B|$

iv $|A \cup B|$

a The numbers up to 100 which are divisible by 3 or 4

b The numbers up to 1000 which are divisible by 10 or 21

c The numbers up to 500 which are divisible by 5 or 8

d The numbers under 100 which are divisible by 6 or 7

e The numbers under 250 which are divisible by 15 or 22

f The numbers under 10 000 which are divisible by 5 or 7

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3 Find $|A \cup B|$ for each of these sets.

a **i** The numbers under 100 which are divisible by 3 or 5

ii The numbers under 1000 which are divisible by 3 or 5

iii Hence, the three digit numbers which are divisible by 3 or 5

b **i** The numbers under 1000 which are divisible by 7 or 10

ii The numbers under 10 000 which are divisible by 7 or 10

iii Hence, the four digit numbers which are divisible by 7 or 10

4 For each of these sets, find:

i $|A|$

ii $|B|$

iii $|C|$

iv $|A \cap B|$

v $|A \cap C|$

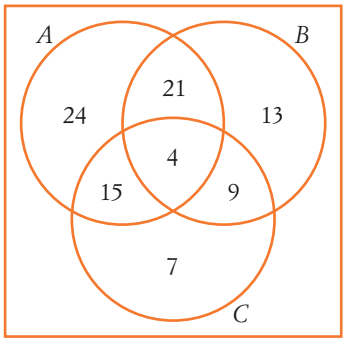
vi $|B \cap C|$

vii $|A \cap B \cap C|$

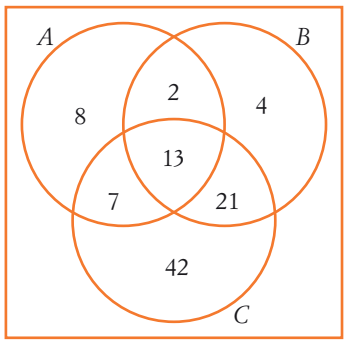
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viii $|A \cup B \cup C|$

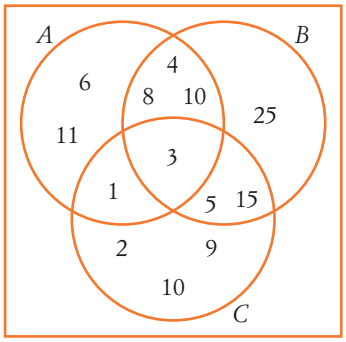
a



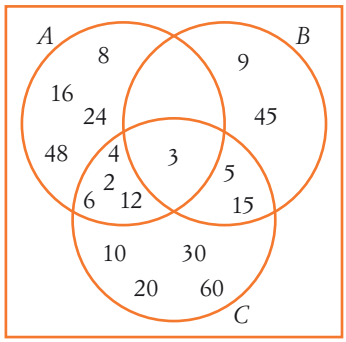
b



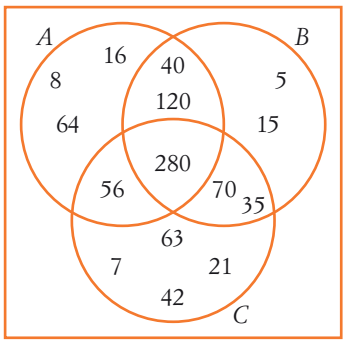
c



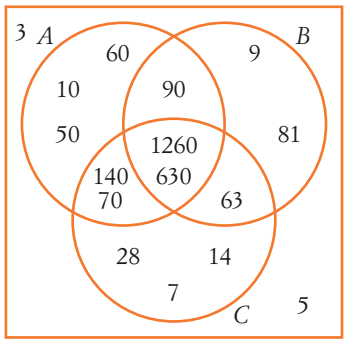
d



e



f



5 For each of these sets, find:

i $|A|$

ii $|B|$

iii $|C|$

iv $|A \cap B|$

v $|A \cap C|$

vi $|B \cap C|$

vii $|A \cap B \cap C|$

viii $|A \cup B \cup C|$

a The numbers up to 100 which are divisible by 3, 5 or 8

b The numbers up to 1000 which are divisible by 2, 5 or 7

c The numbers up to 10 000 which are divisible by 4, 7 or 9

d The numbers under 500 which are divisible by 2, 7 or 11

e The numbers under 5 000 which are divisible by 3, 13 or 23

f The numbers under 7 500 which are divisible by 5, 11 or 17

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6 Find $|A \cup B \cup C|$ for each of these sets.

a i The numbers under 100 which are divisible by 3, 4 or 5

ii The numbers under 1000 which are divisible by 3, 4 or 5

iii Hence, the three digit numbers which are divisible by 3, 4 or 5

b i The numbers under 1000 which are divisible by 13, 17 or 19

ii The numbers under 10 000 which are divisible by 13, 17 or 19

iii Hence, the four digit numbers which are divisible by 13, 17 or 19

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Answers

1	**a**	**i** 2	**ii** 2	**iii** 1	**iv** 3										
	b	**i** 2	**ii** 2	**iii** 1	**iv** 3										
	c	**i** 8	**ii** 5	**iii** 2	**iv** 11										
	d	**i** 9	**ii** 9	**iii** 3	**iv** 15										
	e	**i** 11	**ii** 9	**iii** 3	**iv** 17										
	f	**i** 17	**ii** 14	**iii** 8	**iv** 23										
2	**a**	**i** 33	**ii** 25	**iii** 8	**iv** 50										
	b	**i** 100	**ii** 47	**iii** 4	**iv** 143										
	c	**i** 100	**ii** 62	**iii** 12	**iv** 150										
	d	**i** 16	**ii** 14	**iii** 2	**iv** 28										
	e	**i** 16	**ii** 11	**iii** 0	**iv** 27										
	f	**i** 1999	**ii** 1428	**iii** 285	**iv** 3142										
3	**a**	**i** 46	**ii** 466	**iii** 420											
	b	**i** 228	**ii** 2285	**iii** 2000											
4	**a**	**i** 4	**ii** 4	**iii** 4	**iv** 2	**v** 2	**vi** 2	**vii** 1	**viii** 7						
	b	**i** 4	**ii** 4	**iii** 4	**iv** 2	**v** 2	**vi** 2	**vii** 1	**viii** 7						
	c	**i** 7	**ii** 7	**iii** 7	**iv** 4	**v** 2	**vi** 3	**vii** 1	**viii** 13						
	d	**i** 9	**ii** 5	**iii** 11	**iv** 3	**v** 5	**vi** 3	**vii** 1	**viii** 15						
	e	**i** 7	**ii** 7	**iii** 8	**iv** 3	**v** 2	**vi** 3	**vii** 1	**viii** 15						
	f	**i** 8	**ii** 6	**iii** 8	**iv** 3	**v** 4	**vi** 3	**vii** 2	**viii** 14						
5	**a**	**i** 33	**ii** 20	**iii** 12	**iv** 6	**v** 4	**vi** 2	**vii** 0	**viii** 53						
	b	**i** 500	**ii** 200	**iii** 125	**iv** 100	**v** 62	**vi** 25	**vii** 12	**viii** 650						
	c	**i** 2500	**ii** 1428	**iii** 1111	**iv** 357	**v** 277	**vi** 158	**vii** 39	**viii** 4286						
	d	**i** 249	**ii** 71	**iii** 45	**iv** 35	**v** 22	**vi** 6	**vii** 3	**viii** 305						
	e	**i** 1666	**ii** 384	**iii** 217	**iv** 128	**v** 72	**vi** 16	**vii** 5	**viii** 2056						
	f	**i** 1499	**ii** 681	**iii** 441	**iv** 136	**v** 88	**vi** 40	**vii** 8	**viii** 2365						
6	**a**	**i** 59	**ii** 599	**iii** 540											
	b	**i** 175	**ii** 1770	**iii** 1595											

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